SQL Exercise 4: JOINS, UNION, FILTERING & AGGREGATE

1) SQL Joins

Retrieve all emplayees and their assigned projects using INNER JOIN

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM employees AS A
INNER JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;
```

INNER TABLE JOIN

INNER JOIN

employeeID	firstname	lastname	department	salary	ProjectID	projectname	budget	status
1	John	Doe	IT	70000	101	AI Development	100000	completed
2	Alice	Smith	HR	60000	102	Employee Traning	50000	Ongoing
1	John	Doe	IT	70000	103	CyberSecurity	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	completed
6	Micheal	Clark	Finance	80000	106	Risk Management	80000	Pending

) Retrieve all employees and their assigned project, including
nployees who have no projects using a LEFT JOIN

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
-ROM employees AS A
_EFT JOIN projects AS B
ON A. EmployeeID = B. EmployeeID;
```

Results Table
LEFT JOIN

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
1	John	Doe	IT	70000	103	CyberSecurity	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Micheal	Clark	Finance	80000	106	Risk Management	80000	Pending

)Retrieve all projects and their assigned employees, including
rojects that have no employees using a RIGHT JOIN

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status,
       EmplayeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM Emplayee projects AS A
RIGHT JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;
```

4) Retrieve all emplayees and projects, including those with a match in either table using a FULL OUTER JOIN

```
SELECT EmplayeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM employees AS A
FULL OUTER JOIN projects AS B
ON A.EmplayeeID = B.EmployeeID;
```

Question 2 UNION AND UNION ALL

5) Retrieve a list of all unique cities where employees are located and project statuses.

```
SELECT city AS Location
 FROM employees
  UNION
SELECT status
FROM projects
```

Result (UNIQUE Locations

Location
New York
Los Angele
Toronto
London
Sydney
Completed
Ongoing
Pending

5) Retrieve a list of all cities where employees are located and project statuses, allowing duplicates.

```
SELECT City AS Location
FROM employees
  UNION ALL
SELECT Status AS Location
FROM Projects
```

Result (WITH duplicates)

Location
New York
Los Angeles
Toronto
London
Sydney
New York
Completed
Ongoing
Completed
Pending

Question 3: Filtering Statements

7) Retrieve employees who earn more than 70000

```
SELECT employeeID,
       firstname,
       lastname,
       department,
       salary
FROM employees
WHERE salary > 70000;
```

Results

employeeID	firstname	lastname	department	salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Micheal	Clark	Finance	80000

8) Retrieve employees working in either IT or Finance department

```
SELECT employeeID,
       firstname,
       lastname,
       department,
       salary
FROM employees
WHERE department = 'IT' OR department = 'Finance';
```

Results

employeeID	firstname	lastname	department	salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Micheal	Clark	Finance	80000

)Retrieve projects that are not yet completed

```
SELECT projectID,
       projectname
       budget
       status
FROM   Projects
WHERE  status != 'completed';
```

Results

ProjectID	Projectname	Budget	Status
102	Emplayee Traing	50000	Ongoing
103	Cyber Secourity	75000	Pending
104	Financial Analyst	90000	Ongoing
106	Risk managment	80000	Pending

o) Retrieve projects that have a budget greater than 70000 and not completed

```
SELECT projectID,
       projectname,
       budget,
       status
FROM   projects
WHERE  budget > 70000 AND status != 'completed';
```

Result

ProjectID	Projectname	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

) Retrieve employees from New York OR Toronto order by salary in descending order.

```
SELECT employeeID,
       firstname,
       lastname,
       department,
       salary,
       city
FROM employees
WHERE city ='New York' OR city ='Toronto'
ORDER BY salary DESC;
```

2) Retrieve the top 3 highest paid employees

```
SELECT employeeID,
       firstname,
       lastname,
       department,
       salary
FROM employees
ORDERBY salary DESC
LIMIT 3;
```

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4) Aggregate Functions with GROUPBY and HAVING

13 Find the total salary per department, sorted in descending order

```
SELECT SUM(salary) AS TotalSalary
       department
FROM employees
GROUP BY department
HAVING TotalSalary DESC;
```

Result

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14) Find the average salary per city, but only include cities where the average salary is greater than 65000

```
SELECT city,
       AVG(salary) AS AverageSalary
FROM employees
GROUP BY city
HAVING AverageSalary > 65000
```

Result

City	AverageSalary
New York	75000
Toronto	75000
London	72000

15) Count the number of employees per department, including only department with more than 1 employee.

```
SELECT department
       COUNT(employeeID) AS EmployeeCount
FROM employees
GROUP BY department
HAVING COUNT(employeeID) > 1 ;
```

Departme	EmployeeCount
IT	2
Finance	2

) Retrieve the number of projects per status but only include statuses with at least 2 projects

```
SELECT status,
       COUNT(projectID) AS ProjectCount
FROM   projects
GROUP BY status
HAVING COUNT(ProjectID) >= 2;
```

Results

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

17) Retrieve the total project budget per employee, but only for employe who are managing projects worth more than 150000

```
SELECT employeeID
       firstname
       lastname
       SUM(budget) AS TotalProjectBudget1
FROM   projects
GROUP BY employeeID, firstname, lastname
HAVING SUM(budget) > 150000;
```

Result

EmployeeID	FirstName	lastName	TotalProjectBudget
1	John	Doe	175000